10.3.6 Molecular sieve chromatography

Principle : Gel permeation or molecular size exclusion chromatography is based on the separation of molecules depending on the molecular size and shape. It employs column which contains pores of varying diameter. When the sample is added to the column, molecules of larger size travel faster than the smaller molecules due to permeation into the gel pores. Therefore, larger molecules are eluted first from the column, followed by molecules of smaller size (Fig. 10.6). The gel materials used in gel permeation chromatography are shown in Figure.10.6.

Name	Type	Fraction range (k 0a)
Sephadex G25	Dextran	1-5
Sephadex G100	Dextran	4-150
Sephadex G200	Dextran	5-600
Bio-Gel P30	Polyacrylamide	2.4-40
Bio - Gel P300	Polycrylamide	60-400
Sepharose 6B	Agarose	10-4,000
Sepharose 4B	Agarose	60-20,000

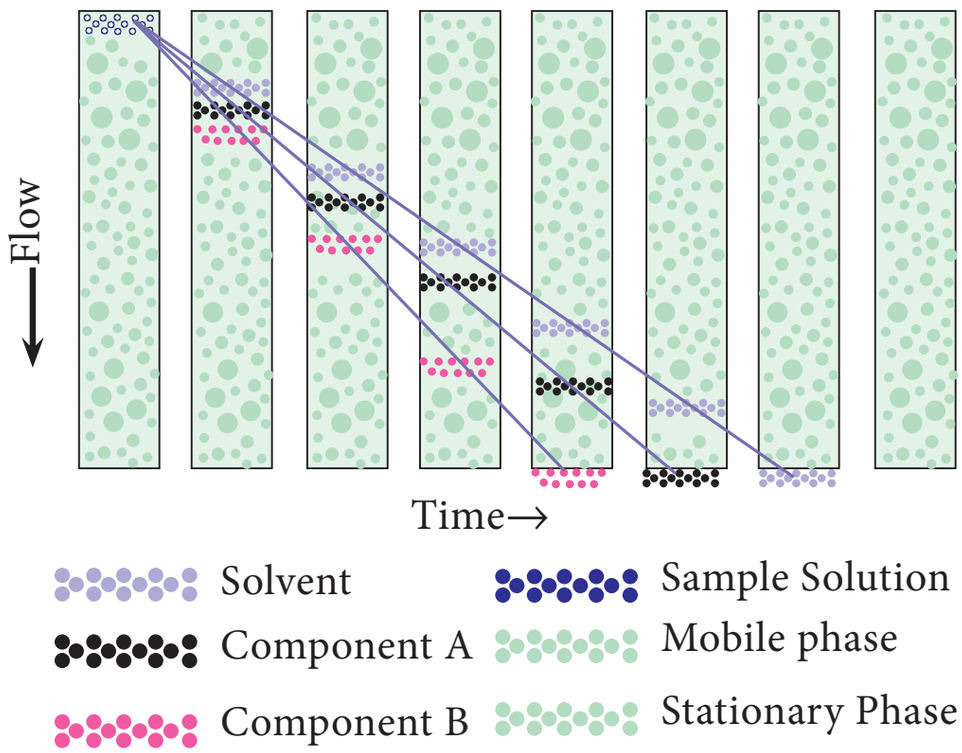


Figure 10.6 : Molecular Sieve Chromatography

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